

Short communication

An investigation of the erosion–corrosion characteristics of ductile cast iron

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Abstract

The erosion–corrosion wear characteristics and mechanism of martensitic ductile cast irons with different alloying compositions in quartz slurries with different pH values were determined and analysed. The results indicated that the erosion–corrosion wear resistance of martensitic ductile cast iron would be increased not only with the pH values of the slurry and the erosion angle, but also with the addition of copper, nickel and chromium elements, respectively. A few of eutectic carbides would appear in the microstructures of ductile cast iron when the chromium element is added into this cast iron. However, these carbides would harm the erosion–corrosion wear resistance of this cast iron in acidic slurry. The addition of Ni, Cu, Cr elements and increase of Si element content would be helpful to improve the erosion–corrosion wear resistance of this cast iron in neutral and alkaline slurry, especially for chromium. Because the hardness of ductile cast iron could be remarkably increased by the occurrence of carbides when the chromium element was added. The erosion–corrosion mechanisms of the martensitic ductile cast irons were dominated by particles erosion wear in alkaline and neutral slurries, and by the interaction of particle erosion wear and medium corrosion in acidic slurry.

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Keywords: Ductile cast iron; Martensitic matrix; Erosion-corrosive; Alloying

1. Introduction

Ductile cast iron, which has excellent mechanical properties, has been widely used as a structural material in the machine, the automobile industry, the mining industry, and so on [1,2]. However, as a wear resistant material, its study and application are still limited [3–6], especially as the slurry pumps material, which is one of the most important equipments in slurry transportation system in ore dressing industry, its application is little or no. The parts of slurry pumps endure the serious erosion of solid particles with a high speed and some corruptions of slurry in ore slurry or fly ash transportation [7–10]. Because the ductile iron has an appropriate

combination of impact fatigue resistance and abrasive wear resistance through the control of its microstructures, as well as lower breakage, lower fatigue spalling and lower production cost, it has been used gradually to make grinding balls and mill liners used under the condition of dry grinding and wet grinding in recent decade, such as that used in cement ball mill and ore dressing ball mill [3,11–15]. The better application effects of ductile cast iron as grinding balls have been obtained. However, the erosion–corrosion wear characteristic of ductile cast iron has hardly been reported. In this paper, therefore, the erosion–corrosion wear characteristics of martensitic ductile iron with different contents of alloying elements are studied under the conditions of different slurries in order to provide the proper material for the producing parts of slurry pumps, as well as to enlarge the application of ductile cast iron in the modern industry.

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2. Experimental procedure

2.1. Material preparation

Ductile cast iron for testing was melted at the temperature of 1550 °C in a 15 kg induction-melting furnace. The melts were poured at a temperature of 1350 °C into 30 mm thick Y-block dry sand molds, and the castings were allowed to cool to room temperature in the molds. Subsequently, the castings were machined into rectangular sample with the sizes of 8 mm × 15 mm × 100 mm. Then, these samples were heat treated for testing at 920 °C for 2 h and quenched in sodium silicate solution to room temperature, and tempered at 250 °C for 2 h. The chemical compositions and hardness of these samples are presented in Table 1. It could be seen that the hardness of the martensitic ductile cast irons increased with the addition of chromium and copper elements, as well as increase of silicon element content in comparison with that of the common ductile cast iron, but the addition of nickel element decreased the hardness of this material to a certain extent. The observation of the material microstructures showed that the microstructures of ductile cast iron were composed of martensite, spherical graphite and a small quantity of retained austenite, as shown in Fig. 1. The eutectic carbides would form with the addition of chromium element, as shown in Fig. 2, thus the hardness of ductile cast iron containing chromium element was the highest. Because the element of Ni would promote the graphitization when the ductile cast iron solidified and would stabilize the austenite when the ductile cast iron was heat treated, the amount of spherical graphite and retained austenite would be increased. This made the hardness of the ductile cast iron decrease after adding nickel element. This microstructure is shown in Fig. 3. When Ni and Cr elements or Cu and Cr elements were added into the ductile cast iron together, the graphitization of Ni or Cu elements would diminish the carbides tendency so that the amount of eutectic carbide would be decreased or rather disappeared, as shown in Fig. 4.

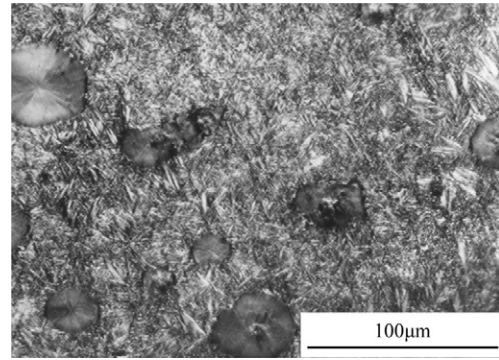


Fig. 1. Microstructure of ductile cast iron.

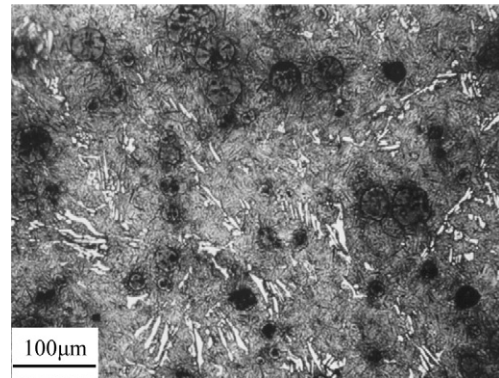


Fig. 2. Microstructures of ductile cast iron with the addition of chromium.

2.2. Erosion–corrosion test

The erosion–corrosion test was carried out with a slurry wear tester as shown in Fig. 5. The tester was mainly composed of a metallic container, sample holder and a stirring blade. Four samples were fixed on the sample holder at the same time and soaked in a slurry when testing. The samples rotated at a radius of 80 mm. The bolts fixed in the holder could adjust the impact angles of solid particles. The slurries were composed of quartz with the sizes of 100–175 μm and

Table 1
Compositions of ductile cast irons for testing

No.	Chemical composition (wt%)									Microstructure	Hardness (HRC)
	C	Si	Mn	Cr	Cu	Ni	S	P	RE		
1	3.62	2.51	0.84				0.025	0.061	0.045	M + G + Ar, Ar < 5%	54.8
2	3.62	2.51	0.85	0.40			0.020	0.063	0.053	M + G + 5–10%K	59.2
3	3.62	2.51	0.86		0.50		0.023	0.060	0.057	M + G + Ar, Ar < 5%	55.8
4	3.62	2.51	0.83			0.50	0.025	0.059	0.049	M + G + Ar, Ar < 5%	51.8
5	3.62	2.51	0.85	0.40		0.50	0.020	0.057	0.053	M + G + Ar, Ar < 5%	56.3
6	3.62	2.51	0.87	0.40	0.50		0.020	0.062	0.048	M + G + Ar, Ar < 5%	58.3
7	3.62	3.05	0.84				0.024	0.063	0.047	M + G + Ar, Ar < 5%	54.8
8	3.62	3.52	0.85				0.021	0.060	0.049	M + G + Ar, Ar < 5%	54.6

Note. M – martensite, Ar – retained austenite, G – spheroidal graphite, F – ferrite, K – eutectic carbides. The amount of retained austenite in ductile cast iron was determined by X-ray diffraction analysis.

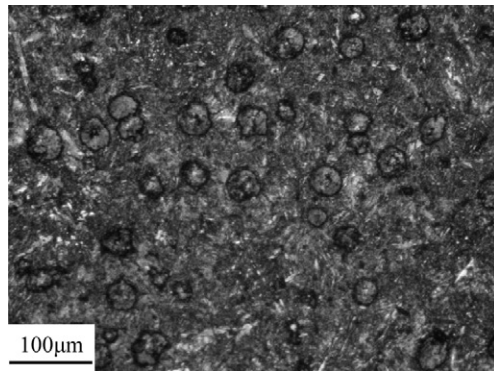


Fig. 3. Microstructures of ductile cast iron with the addition of nickel.

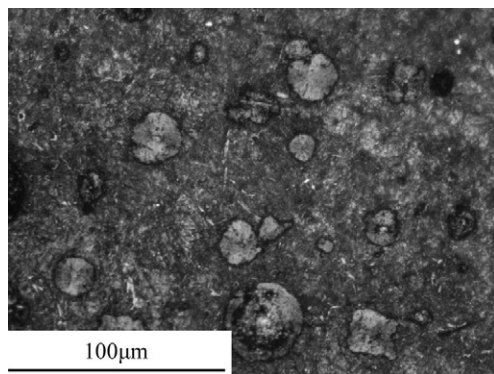


Fig. 4. Microstructures of ductile cast iron with the addition of nickel and chromium.

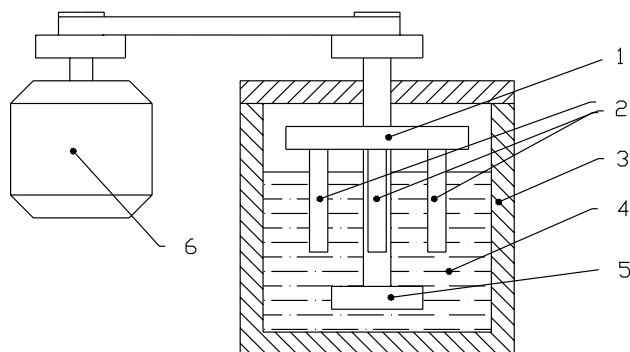


Fig. 5. Schematic diagram of the erosion-corrosion tester: (1) sample holder, (2) samples, (3) slurry pool, (4) slurry, (5) stirring blade, (6) motor.

sulphuric acid solution with pH 3, water with pH 7 and sodium hydroxide solution with pH 12, respectively. The ratio of quartz and water in slurry was 3:4. When the test was carried out, the motor drove the samples rotary in the slurry, and the samples speed was 5.86 m/s. The surfaces of the samples were impacted by the quartz particles in the slurry and corroded by the solutions with different pH values. The impacting angle of quartz particles on the surface of samples was 45° and 90°, respectively. After 30 min for the running-in test, the slurry was tipped out from the slurry pool and new slurry

was poured into the slurry pool. The three tests were performed on each sample with different compositions, and each test running for 30 min. The wear resistance was taken as the reciprocal of the weight loss, i.e.,

$$\varepsilon = \frac{1}{\text{weight loss}}.$$

A balance with a sensitivity of 0.0001 g was used to determine all of the weight losses of samples in the tests.

The wear surface morphology was observed in the scanning electron microscope of Amary-1000B to study the mechanism of erosion-corrosion wear of ductile cast iron in erosion-corrosion wear tests.

The corrosion potential of ductile casts was determined by potentiostat. The corrosion media were sulfuric acidic solution with pH 2 and 4, water with pH 7, and sodium hydroxide solution with pH 10 and 12, respectively.

3. Results and discussion

The erosion-corrosion wear resistances of ductile cast irons with different compositions under the conditions of different impact angles and pH value slurries are shown in Figs. 6 and 7. It could be seen that the wear resistance of ductile cast irons would be increased with the increase of the impact angle of solid particles, pH value of slurries and content of silicon element, as well as with the addition of Cr, Ni, Cu elements. But wear resistance of ductile cast iron would be decreased in acidic slurry when the Cr element was added into the ductile cast iron. This indicates that the erosion-corrosion wear resistance of ductile cast iron is the synthetic expression of its physical and chemical characteristics under definite test conditions, i.e., systematic properties, but is not its inherent characteristic of the testing ductile cast iron. In other words, erosion-corrosion wear of ductile cast iron is determined by the summing up of

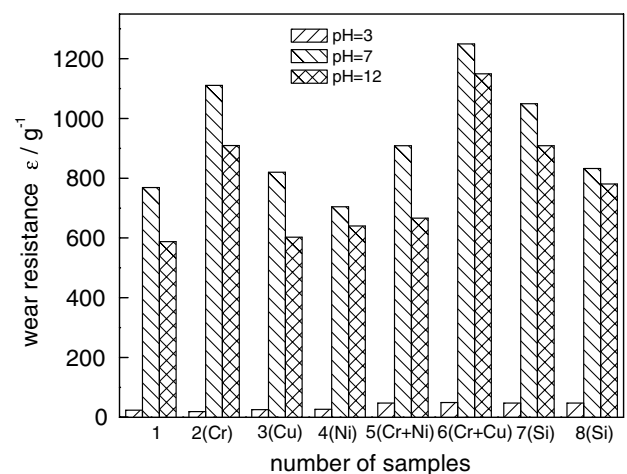


Fig. 6. Wear resistance of ductile cast iron with different compositions eroded by different pH value slurries at the impact angle of 90°.

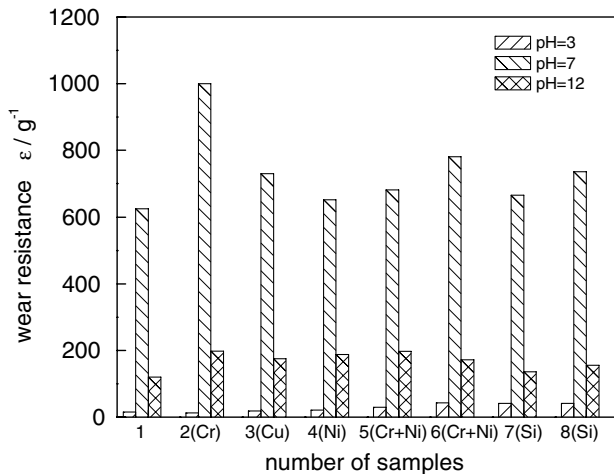


Fig. 7. Wear resistance of ductile cast iron with different compositions eroded by different pH value slurries at the impact angle of 45°.

all the processing appearing in the test, and its occurrence and development depends upon both the properties of ductile cast iron and the external test conditions.

3.1. Effect of impact angle

The change of impact angle would change the weight loss means of tested materials when the abrasive particles in the slurry impacted on their surfaces, so that it

would affect the erosion–corrosion wear resistance of tested materials. When the particles in slurry impacted on the surface of samples at 90°, the impingement pits and plastic deformations would be produced on the surfaces of samples. At the same time, numerous particles impacted on the surfaces of samples making the surfaces plastic deformed again and again and producing the fatigue cracks. When the fatigue cracks propagated, the fatigue spalling would be produced, as shown in Fig. 8. So the erosion–corrosion wear mechanisms of ductile cast iron were the impact pits and plastic deformation fatigue spalling at the impact angle of 90°. However, when the impact angle of solid particles was 45°, the particles impacted on the surface of samples and slipped on the surface of samples. This led to the cutting of the surface of samples and the loss of surface materials, as shown in Fig. 9. At this time, the wear mechanism of ductile cast iron was dominated by the micro-cutting. Therefore, the erosion–corrosion wear resistance of ductile cast iron was lower at the impact angle of 45°, i.e., the erosion–corrosion wear weight losses of tested materials were larger at the impact angle of 45°.

3.2. Effect of pH value of slurries

The pH value of slurry media would affect the erosion–corrosion wear resistance of ductile cast iron except the particles impact angle, as shown in Figs. 6

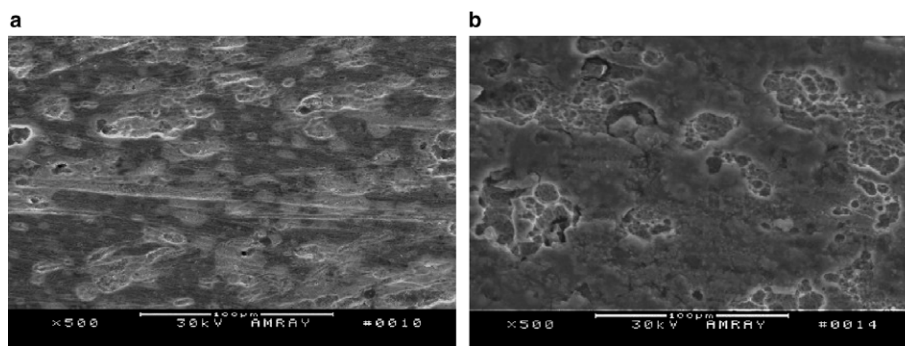


Fig. 8. Erosion–corrosion wear surface morphology of ductile cast iron with 3.5% Si at the impact angle 90° in: (a) alkaline slurry and (b) acidic slurry.

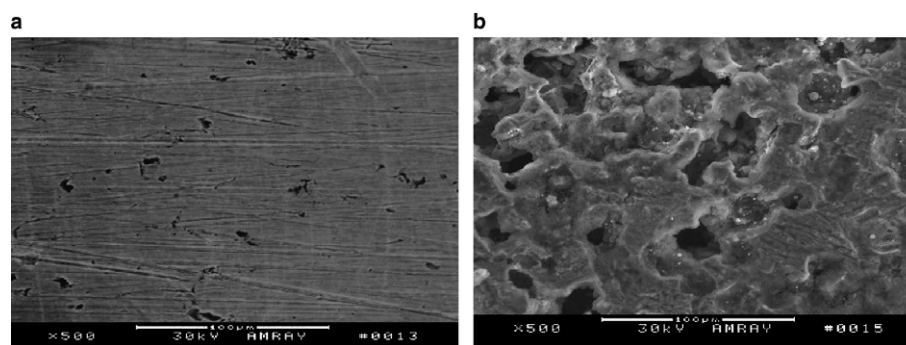


Fig. 9. Erosion–corrosion wear surface morphology of ductile cast iron with 3.5% Si at the impact angle 45° in: (a) alkaline slurry and (b) acidic slurry.

and 7. The scanning electron microscope (SEM) observation of wear surface showed that the erosion–corrosion wear mechanisms of ductile cast iron were dominated by micro-cutting of particles and fatigue spalling in the test conditions of neutral and alkaline media, but the corrosion action of media would hardly be seen, as shown in Figs. 8(a) and 9(a). The corrosion potential of ductile cast iron is also higher in alkaline solution, as shown in Fig. 10. Therefore, the erosion–corrosion wear resistance of ductile cast iron would be raised sometimes. However, under the test conditions of acidic slurries, the appearance of grain boundary corrosion which is caused by electrochemistry corrosion would split the matrix of ductile cast iron and decrease its strength so that the wear of ductile cast iron would be accelerated in acidic medium, as shown in Figs. 8(b), 9(b) and 11. The corrosion potential of ductile cast iron in acidic solution is also lower, as shown in Fig. 10. Thus, under the conditions of acidic slurries, the corrosion would accelerate the wear of ductile cast iron. At the same time, the solid particles also led to the loss of the passivated surface of ductile cast iron caused by cor-

rosion of acidic medium. This would accelerate the ductile cast iron corrosion again [16]. Thus, the erosion–corrosion wear resistance of ductile cast iron was lower in acidic slurries because of the interaction of acidic medium corrosion and abrasive particles impact wears.

3.3. Effect of alloy elements

The hardenability of ductile cast iron could be increased when the alloying elements were added into it, but the effects of different alloying elements added in ductile cast iron on its microstructures and hardness were different, as shown in Table 1. According to our previous research results, the amount of carbide would increase to about 10% in ductile cast iron when 0.5% chromium was added into it [17]. Thus, the hardness of ductile cast iron would be increased by the appearance of carbide in its microstructures and the strengthening of chromium element solid solution. Because the erosion–corrosion wear mechanism of the ductile cast iron was dominated by micro-cutting of particles at impact angle 45°, and by impact pits and fatigue spalling at impact angle 90° in neutral and alkaline media, its erosion–corrosion wear was dependent on its hardness and fatigue strength. Thus, the wear resistance of ductile cast iron would be increased with the increase of its hardness and fatigue strength when chromium element was added into it. However, in the acidic slurries, because acidic media corroded the phase interfaces between the metallic matrix and carbide and made the carbide in the microstructures of ductile cast iron lose the support of metallic matrix, the carbide would be broken off and dropped out by the impacts of particles. At the same time, the metallic matrices also lose the protection of carbide and were cut out easily by solid particles. Thus, the wear resistance of ductile cast iron containing chromium element was not the best in the acidic medium. When the chromium and copper or chromium and nickel elements were added into the ductile cast iron together, the amount of carbide precipitated during the solidification of ductile cast iron would be decreased because of the graphitization of copper and nickel elements. The phase interfaces corrosion would be weakened. The solid solution strengthening of chromium and copper or chromium and nickel elements made the hardness of ductile cast iron increase. Therefore, the erosion–corrosion wear resistances of ductile cast iron containing the chromium and copper or chromium and nickel elements were the best.

Though the hardness of ductile cast iron in which was added the nickel element was lower than that of other elements, its erosion–corrosion wear resistance was better in the acidic medium. This was because of the fact that the nickel element increased the corrosion potential of metallic matrix of ductile cast iron except

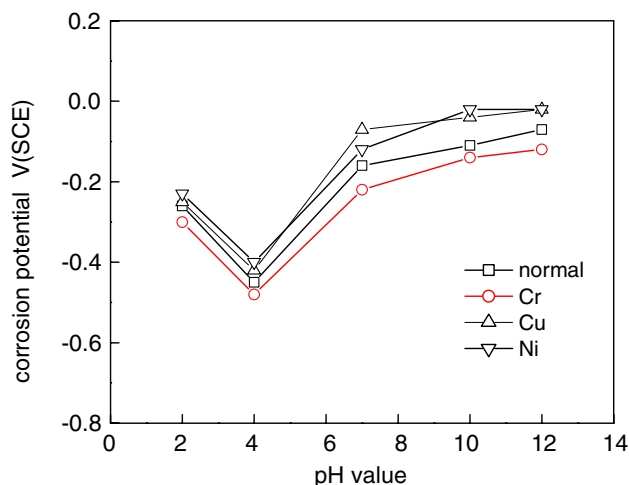


Fig. 10. Relationship of corrosion potential, media pH values and addition of alloying elements.

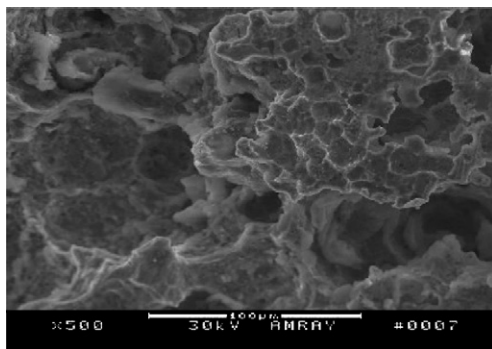


Fig. 11. Erosion–corrosion wear surface morphology of ductile cast iron at the impact angle 90° in acidic slurry.

for increasing its hardenability, as shown in Fig. 10. The corrosion rate of ductile cast iron would be decreased on acidic medium. The copper and silicon elements had the same action as the nickel element when they were added into the ductile cast iron. Therefore, these elements could increase the erosion–corrosion wear resistance of ductile cast iron when they were added into the ductile cast iron.

4. Conclusions

- (1) The erosion–corrosion wear resistance of ductile cast iron would be increased with the increase of the pH values of the slurries and the impact angle of solid particles.
- (2) The wear mechanism of the ductile cast iron was dominated by the combination of solid particles micro-cutting wear and medium corrosion in acidic slurry, and by solid particles micro-cutting wear in neutral and alkaline slurry, compared with a little corrosion.
- (3) The erosion–corrosion wear resistance of ductile cast iron would be increased with the addition of Ni, Cu, elements, increase of the silicon content in acidic slurry, and by the addition of Cr, Ni, Cu elements in neutral and alkaline medium slurry, respectively.

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